Creating Value with MBSE in the High-Tech Equipment Industry

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■ **ABSTRACT**
The Netherlands has a strong presence in the high-tech equipment industry sector with world-wide renowned organizations. Systems engineering is a key capability that is well-established in this sector. The industry now sees model-based systems engineering (MBSE) as indispensable to bringing systems engineering capabilities to the next level. Despite this, MBSE is not a fully established practice in this sector as in other industries. ESI has initiated a study to understand the background of the sector's interest in MBSE, the challenges to address with MBSE, experiences with, and fit of current MBSE methodologies versus the characteristics of this sector. This article reports on the results of this study. It highlights innovation in MBSE to address the needs and characteristics of the high-tech equipment industry.
■ **KEYWORDS:** MBSE, high-tech equipment industry, digital transformation, brownfield development, systems engineering

SYSTEMS ENGINEERING FOR THE HIGH-TECH EQUIPMENT INDUSTRY

The Netherlands has a strong presence in the high-tech equipment industry sector with world-wide renowned organizations. Innovations now take these systems (for example, nanometer-accurate lithography systems, angstrom resolution electron microscopes, minimally invasive medical equipment, commercial printing equipment, and advanced warehousing systems) towards unprecedented levels of features and functions, increasing complexity every day. Consequently, R&D organizations have grown, with (business-) critical issues needing to be addressed in the ecosystems of partners (supply chain partners, field service partners, innovation partners).

Some characteristics of this industry sector are the annual production quantities, typically 10s to 1000s, while offering an extensive product portfolio with product variants, options, and sometimes customer-specific features. This diversity makes almost any delivered product unique. The organizations operate in commercial markets; some industries experience strong competition, while others are unique suppliers or market leaders. To deal with this business context, these industries apply an evolutionary way of working for their R&D. Solutions are developed incrementally, using the previous product as a baseline:

- The industry has established product families supported by product platforms to deal with the variety. Product innovations should comply with platform thinking.
- The products add market value to system qualities such as performance, throughput, or uptime, which requires a robust system-level architecting approach.
- The industry has widely adopted an agile way of working, which fits well with the evolutionary way of working.
- The products and solutions offered to the market have a long lifetime (often well over twenty years), resulting in a large installed base. It has a key business value because it shows its leading position in the market and provides opportunities for upgrade and replacement sales for a service business.
- The industries are traditionally strong physics and electro-mechanics oriented. However, in the past decades, they have experienced an ever-increasing effort in software development.
- Further, the solutions used to be standalone applications but require integration into a larger system of systems.
- Design knowledge is captured in documents, which are sometimes generated from databases.

Many R&D employees are employed for a lengthy period; sometimes, they work their whole professional career at a single company. They have in-depth knowledge about current developments as well as the installed base. Although this knowledge is essential, keeping it up to date is expensive.

The technical and business complexity forces these industries to grow, which means an influx of new people — who do not have the complete design history in their minds. Also, the retirement of senior employees working on crucial expertise is a source of loss of know-how. A solution for retaining critical know-how is vital to maintain its market position.

The above drivers cause this industry sector to have an increasing interest in replacing its classical systems engineering approach with a (more) model-based approach. ESI and the sector hence started a study (Wesselius, van den Aker, et al. 2022) to see what MBSE can bring to the sector, the challenges to address with MBSE, experiences with, and to gauge the fit of current MBSE methodologies versus the characteristics of the high-tech equipment industry.

MBSE VERSUS MODELS IN SYSTEMS ENGINEERING

Systems engineering is both about technical engineering aspects and engineering management: ensuring that all engineering is done for system effectiveness in a controlled way. Models abound in systems engineering these days. Are all system engineers doing MBSE, at least to some extent? The answer is "no." MBSE is not about "using models while systems engineering;" instead, the fundamental MBSE tenet is that models become the primary asset, the authoritative source of information for everyone. In MBSE, models are not add-ons to (authoritative) documents. Instead, models replace those documents. If documents are needed, they are generated from the models. In case of doubt, the models are authoritative and overrule the documents.

In the Dutch high-tech equipment industry, models abound, but they are not yet the authoritative source of information. Most systems engineering related models have a single purpose and are disconnected. The sector, however, looks to MBSE to improve its systems engineering practice. A Sandia report (Carroll and Malins 2016) and the MBSE Survey of Stevens Institute of Technology (Cloutier and Bone 2015) show positive results in terms of cost savings and quality improvements for organizations that adopted MBSE. To introduce MBSE successfully, the Sandia report also names several prerequisites that need to be in place: related to systems engineering processes, training of system engineers, and investment in full-scale MBSE tools. Also, the organization needs to commit to processes, resources, people, and infrastructure to support model management throughout the system design life-cycle.

Both these reports strongly focus on the aerospace and defense industry. Commercial, high-tech equipment manufacturers have a different business context, influencing the role and value of systems engineering and MBSE. Therefore, the MBSE experiences described in these reports cannot be carried over one-on-one to the commercial sector. Understood must be which are the main drivers for MBSE in this sector.

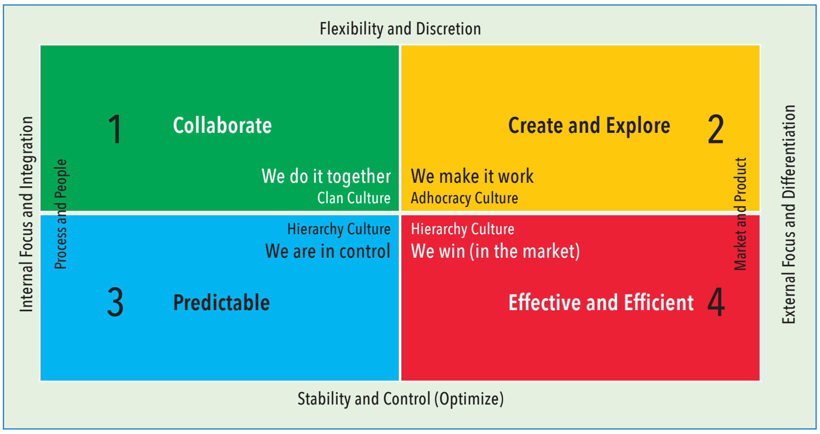

Figure 1. Quadrants to map motivations for MBSE introduction (inspired by Quinn and Cameron and Insights Discovery)

DRIVERS FOR MBSE ADOPTION IN THE HIGH-TECH EQUIPMENT INDUSTRY

Motivations for MBSE have typically been expressed in organization-specific terminology. To analyze these drivers uniformly, we used a four-quadrant matrix inspired by the competing values culture model (Cameron en Quinn 2006, Quinn Association 2022) and the insights discovery color coding (The Insights Group Limited 2021).

Each quadrant captures a class of MBSE drivers which may be related to the culture and business strategies of the organizations. These quadrants (see Figure 1) are created by combining two axes: (1) internal focus and integration or external focus and differentiation, (2) stability and control or flexibility and discretion. The first axis looks at how an organization operates in the market: based on an internal focus and integration or an external focus and differentiation versus the competition. The second axis looks at how an organization approaches the effectiveness of its workforce and work processes: via control and stability (and strict processes) or flexibility and discretion. This yields four quadrants as follows:

1. **Collaborate [Internal Focus and Integration | Flexibility and Discretion]**
   This quadrant focuses on drivers for collaboration aspects. Motivators are positioned as follows:
   - Having independent teams doing concurrent engineering, effective collaboration ensures that their design deliverables integrate into a product.
   - The need to spread system knowledge throughout the organization.
   - The desire to enhance collaboration across disciplines in self-managing teams.
2. **Create and Explore [External Focus and Differentiation | Flexibility and Discretion]**
   This quadrant focuses on drivers for conceptualization, exploration, and trade studies. Motivators are positioned as follows:
   - Exploring design options and simulating their consequences at a system level.
   - Performing trade-space analysis and make trade-off decisions, taking the system-wide scope into account.
3. **Predictable [Internal Focus and Integration | Stability and Control]**
   This quadrant focuses on drivers for keeping control and consistency. Motivators are positioned as follows:
   - Assuring that all requirements are met and verified.
   - Standardizing the way of working throughout the organization.
4. **Effective and Efficient [External Focus and Differentiation | Stability and Control]**
   This quadrant focuses on drivers for bringing products fast to market. Motivators are positioned as follows:
   - Being able to quickly compose customer-specific systems from platforms of pre-designed and pre-released components.
   - Reduce the time from ordering to delivering a system to the customer.
   - Being able to ensure properties of multiple system configurations.

MAPPING MBSE DRIVERS TO THE NEEDS OF THE HIGH-TECH EQUIPMENT INDUSTRY

In the MBSE study of ESI and Partners, (Wesselius et al. 2022), the high-tech equipment industry sector put forward four main drivers to explore the introduction of MBSE:

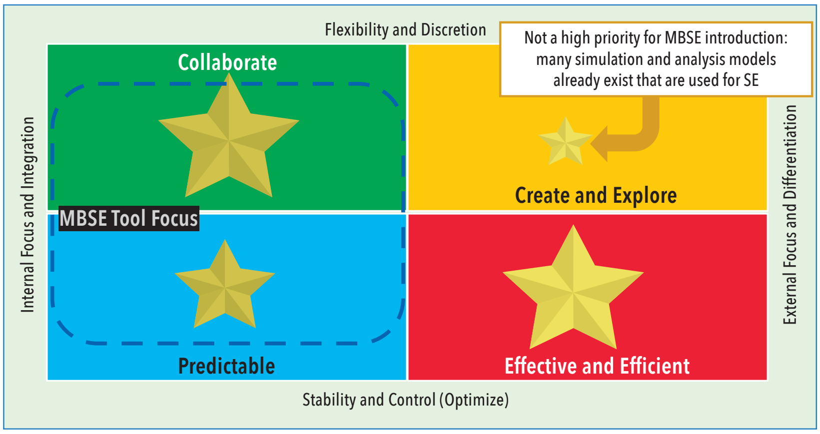

Figure 2. Relative importance of motivations for MBSE in the Dutch high-tech industry

1. Enhancing cooperation and knowledge exchange (quadrant 1).
2. Consolidating knowledge from legacy systems for new generations of engineers (quadrant 1).
3. Leveraging platforms to accelerate system/solution design and delivery (quadrant 4).
4. Assuring the properties/qualities of system variants shipped to customers (quadrant 4).

Figure 2 shows the ranking of MBSE motivations expressed by the Dutch ecosystem, overlaid on the focus of current MBSE tools: quadrant 3 (predictable) while supporting quadrant 1 (collaborate). quadrant 2 (create and explore) surprisingly scored significantly lower with the Dutch ecosystem. This could create the impression that their interests lay outside of using models to speed up innovation. This conclusion, however, is incorrect as they already use models and tools for this purpose. These (engineering) models are typically not connected and do not establish an "authoritative source of truth." However, they expect MBSE to enable smoother communication and knowledge exchange about those innovations, which is part of the drivers in quadrant 1 (collaborate).

Furthermore, all industries expect the core benefits of MBSE: the authoritative, consistent, and easily accessible system-wide information (quadrant 3). To them, this is the enabler that MBSE is expected to provide.

GREENFIELD VERSUS BROWNFIELD SYSTEMS ENGINEERING

The evolutionary way of working, that is the *brownfield* business context of the Dutch high-tech equipment industry, is important. "Brownfield" can be characterized by: (i) this year's system is an incremental, evolutionary innovation of last year's system; (ii) a substantial part of the business is related to legacy systems in their installed base (iii) therefore, a large portion of their R&D capacity is used to sustain their installed base, including the delivery of upgrades to systems that are twenty to thirty years old. The brownfield nature of this sector's business context gives specific challenges that also drive their motivations for exploring MBSE:

- Organizations must efficiently and effectively sustain their installed base for long periods, so they need teams knowledgeable of those systems. Developing system-level domain knowledge for new engineers requires considerable time and a steep learning curve. Therefore, consolidating systems engineering knowledge of legacy systems and making it easily accessible to new engineers is crucial for their business. This is a motivator for quadrant 1.
- As they have delivered systems in many configurations and generations, consolidating the knowledge required to sustain the installed base is complex. MBSE is hoped to bring an innovative way to structure the required systems engineering information and assure consistency for systems in the field.
- As next year's system is, in most cases, an evolutionary innovation from this year's system, in-depth knowledge about this year's system is crucial for successful innovation. This also puts a strong focus on knowledge consolidation and knowledge exchange.

What complicates the introduction of MBSE in such *brownfield* organizations is that there are no authoritative models for current or past systems. Instead, large sets of documents are available, typically neither fully up-to-date nor entirely consistent. Next to these documents, the knowledge is typically captured in organizations; processes; databases; people (the experienced "local heroes"). Introducing MBSE requires leap-frogging this chasm. A complex transition path is needed in which hybrid approaches (document-based and model-based systems engineering) will co-exist.

POSITIONING METHODS AND TOOLS ON INDUSTRY MBSE MOTIVATORS

The MBSE study of ESI and partners (Wesselius et al. 2022) was conducted as a series of workshops with the high-tech equipment industry, where in turn, every partner explained their MBSE ongoing work as the basis for discussion. In addition, interviews with leading MBSE tool vendors were conducted to explore the capabilities and strategies of contemporary MBSE methods and tools. Mapping our quadrant observations leads to the following overview (Figure 3).

- Quadrant 1: MBSE tools to collaborate [Green = Internal | Flexible]
  Several tools provide advanced collaboration environments supporting distributed teams. In this quadrant, we did not find methods or tools that actively

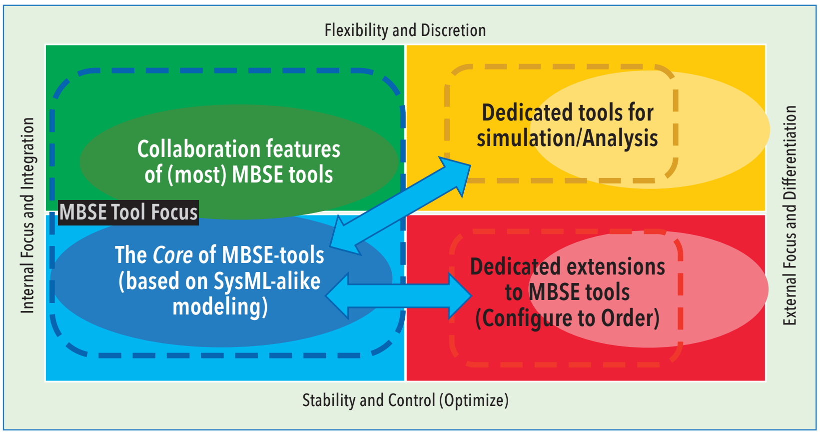

Figure 3. Positioning methods and tools relative to MBSE motivations.

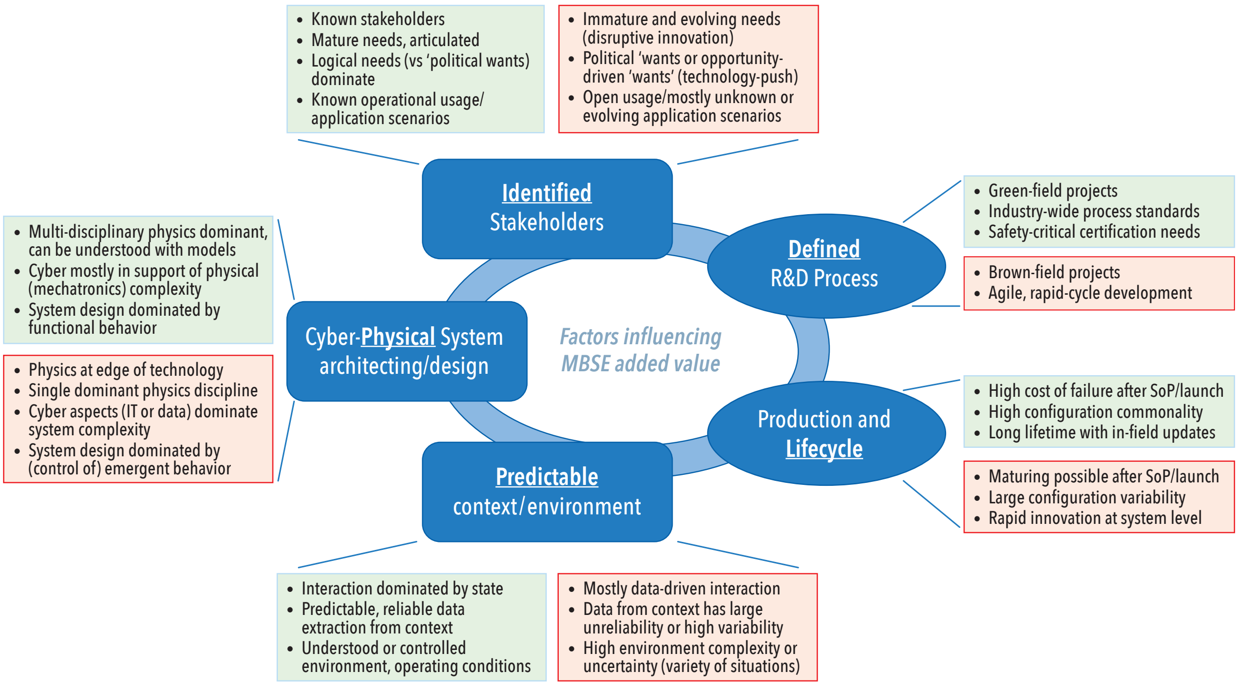

Figure 4. Factors influencing the added value of MBSE (positive factors in green, negative factors in red)

support working in an evolutionary brownfield environment and capturing legacy and current systems' design rationale or intent.

- Quadrant 2: MBSE tools to create and explore [Yellow = External |Flexible]
  Modeling is a key aspect of design exploration and analysis. Many dedicated methods, tools, and techniques are available to analyze and simulate specific system aspects. Strong interfaces between dedicated tools and MBSE-core tools are needed at a system level to maintain a consistent, authoritative source of systems engineering information.
- Quadrant 3: Predictable core of MBSE-tools [Blue = Internal | Control]
  Most MBSE tools focus on quadrant three by providing SysML-alike modeling languages and techniques to create RFLP-alike structures: In this quadrant, MBSE tools primarily offer techniques for function decomposition and allocation but less to system qualities and system properties (a next-generation SysML V2 (OMG 2022) is in the works to address these issues).
- Quadrant 4: Tools to be efficient and effective in the market [Red = External | Control]
  Several MBSE vendors have the strategy to deliver complete and integrated tool suites, including product lifecycle management and manufacturing. Others focus on MBSE-core functionality and have the strategy to connect to dedicated PLM, simulation, and analysis tools through the open interfaces provided by these.

During the study, several more generic observations and attention points were identified. Firstly, modeling is a crucial aspect of design exploration and analysis (quadrant 2). This needs strong interfaces between simulation and analysis tools and the MBSE-core tools (quadrant 3) to assure consistency, cohesion, and authoritativeness to support collaboration and concurrent engineering (quadrant 1). Secondly, suppose the models become the authoritative source of systems engineering information. In that case, they should also capture design rationale and intent (today, architects spend much time talking with design teams to convey these). Thirdly, MBSE methods and tools are needed to create models from legacy design artifacts (documents, Excel sheets, Visio diagrams, CAD files) in a brownfield environment. Lastly, given the sector's MBSE motivations, integrating key aspects of platform-thinking and product line engineering into the MBSE-core methods and tools is needed for MBSE to be effective in this sector, including reasoning about system variants/diversity and across legacy.

CONSIDERATIONS FOR INTRODUCING MBSE IN THE HIGH-TECH EQUIPMENT INDUSTRY

MBSE has been first applied in aerospace and defense in long-running "engineer-to-order" type projects. From then on, other domains have adopted or experimented with MBSE. How do these organizations and experiments relate to the characteristics of the high-tech equipment industry?

Factors Influencing the Added Value of MBSE Introduction

To support the high-tech equipment industry in introducing MBSE, influencing factors were identified for the likelihood of added value for MBSE over "just" doing systems engineering. These factors were inferred from success reports of the application of MBSE in various domains and complemented with insight into the nature and strengths of MBSE methods and tools. Figure 4 presents an overview of these (generic) influencing factors.

As shown in Figure 4 (on the left), the nature of the systems may have a considerable influence on whether MBSE could add significant value. MBSE thrives when the design challenge is balancing multi-disciplinary physics (hence the underlined <u>physical</u>). When cyber aspects or management of emergent behavior dominates complexity, MBSE is less suitable for managing such aspects.

Stakeholders should be known and able to articulate their needs (hence the underlined <u>identified</u> in Figure 4). MBSE needs well-articulated system requirements, which form the basis of traceability into the

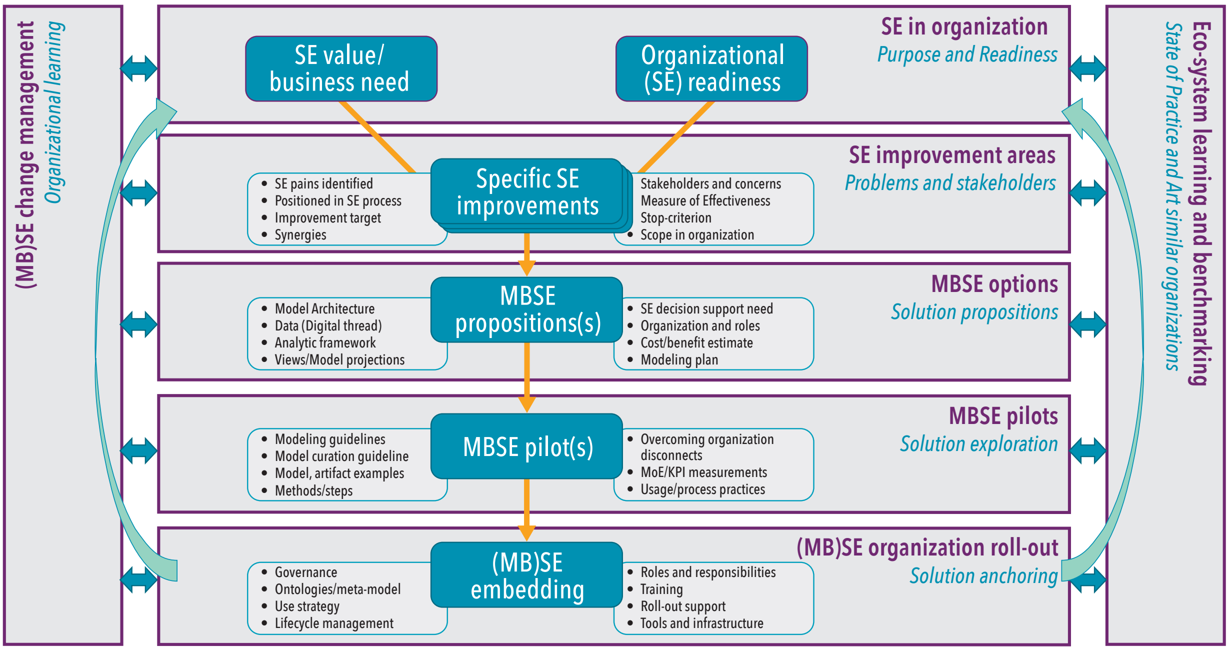

Figure 5. Reasoning line to guide the introduction of MBSE in an organization

design and verification of its decomposition and properties.

A system's context (Figure 4 bottom) should be understood and captured in models (hence the underlined predictable). When this context sees considerable uncertainty or mostly data-driven interaction, then this presents difficulties for (typically function-oriented) MBSE approaches. Specialized approaches are then called for (for example, capturing driving scenarios in automated driving).

Concerning the R&D process (Figure 4, right and top), greenfield projects allow the complete use of MBSE over the full design scope (emphasized by the underlined defined). In contrast, brownfield development for incremental design upgrades faces missing models and lost know-how. Recreating these for MBSE incurs high R&D overhead and long time-to-value.

Finally, concerning production and lifecycle (Figure 4, right and bottom), MBSE is particularly suited to minimize unacceptable risks when a high cost of failure could occur after start-of-production (SoP) (indicated by the underlined lifecycle) as is typically the case with safety-critical systems, such as road vehicles or commercial aircraft. For products that can be launched quickly as minimally viable products, MBSE overhead may be too much.

Most MBSE methods and tools implement a variant of the requirements-functional-logical-physical (RFLP) approach. This approach has been well-suited for particular domains and organizations. Figure 4 provides contrasting factors to consider where MBSE could add the most value or where the value could be less significant than just doing Systems Engineering with (disconnected) models.

A Reasoning Line for the Introduction of MBSE in the High-Tech Equipment Industry

The introduction of MBSE, as part of an organization's digital transformation, requires "an integrated digital approach that uses authoritative sources of system data and models as a continuum across disciplines to support lifecycle activities from concept through disposal" (Department of Defense – Office of the Deputy Assistant Secretary of Defense for Systems Engineering 2018). Besides introducing tools and methods, much attention and effort must be spent on creating organizational capabilities. To achieve the full benefits of digitizing (systems) engineering, just introducing MBSE tools and the associated methods does not suffice.

Thus, how to guide (and anchor) an introduction of MBSE in an organization? Achieving/perceiving value is far from trivial (Cloutier and Obiako, 2020). ESI created a reasoning line for this purpose (see Figure 5), which considers both technical and organizational aspects for MBSE to add value and needs to be embedded in a Systems Engineering way-of-working.

This reasoning line considers seven main viewpoints to guide, customize, and rationalize a value-add introduction to MBSE, and are annotated with relevant technical and organizational aspects to be considered.

The first viewpoint looks at the state of **systems engineering in the organization**: its systems engineering readiness and the business need for change. An organization must be capable of doing systems engineering before embarking on MBSE. Also, the intended change must be rooted in a clear business need/value improvement.

Secondly, **systems engineering improvement area**s should be understood and identified, the "problem space" (Noguchi, Minnichelli, and Wheaton 2019). What are the systems engineering pains? Which stakeholders experience these? What part of systems engineering needs to be improved, how much, when to stop (and when is good, good enough)? Also, which part(s) of the organization should be involved? Do the outcomes address the real needs of "outside" beneficiary stakeholders (sales, service, lifecycle)?

Thirdly, based on rationalized and scoped Systems Engineering improvements, a selection can be made where **MBSE options** could add value. For those, value propositions should be defined, including the systems engineering support targeted and how to achieve this with MBSE (which model(s) include analytic framework and data), but also how to organize and plan this, with a cost/benefit analysis. An overview of potential MBSE value options (benefits) is given in (McDermott et al. 2020).

Fourthly, **MBSE pilots** can explore these options' effectiveness and measure/assess benefits. Pilots also can refine methods and provide (input for) guidelines. Pilots may encounter organizational issues and disconnects exposed by a more formal way of working.

Fifthly, **(MB)SE organization roll-out** needs to ensure that the MBSE way of working is sustained by embedding it in the organization. This requires governance, ontologies/meta-models, tools, infrastructure, training, and definition of (new or changed) roles and responsibilities.

Two further supports for the activities include i) general **(MB)SE change management** to ensure that organizational learning and a change in Systems Engineering culture takes place, and ii) **ecosystem learning / benchmarking** to ensure lessons learned in similar organizations are incorporated, not duplicated.

The introduction of MBSE is a complex change process affecting many parts of an organization and how they collaborate. The reasoning line aims to guide MBSE introduction in the high-tech equipment industry but has wider applicability. Its purpose is a check for the rationalization of activities: not to be mistaken for a process (Parnas and Clements 1986). Having an articulated purpose and rationalization of MBSE activities is crucial to gain organizational support, achieving value, and for MBSE to be firmly embedded in Systems Engineering.

CONCLUSION AND OUTLOOK

In the high-tech equipment industry sector, invariably development of a new system starts from the design of the earlier generation of systems. These organizations typically support an installed base with systems up to fifteen to thirty years old. On the one hand, this installed base is a key business value; it shows their leading position in the market and provides opportunities for upgrade and replacement sales as a service business. On the other hand, supporting an aging installed base is costly and requires their teams to know past system generations. Furthermore, current development is increasingly software intensive and of increasing multi-disciplinary complexity.

Like the aerospace and defense industry, the high-tech equipment industry thus needs the consistency and completeness promised by MBSE ("authoritative source of truth") to battle this complexity and needs the cooperation/collaboration platforms that most MBSE tools offer. In addition, this industry also needs ways to leverage the value of MBSE in their business environment, characterized by dealing with brownfield R&D, evolutionary delivery requiring knowledge about past system generations, and leveraging investments in platforms.

These needs expose a gap in what MBSE currently offers and impose conditions for successful MBSE usage in the high-tech equipment industry (see also (Wesselius, MBSE for the High-Tech Equipment Industry - MBSE-study of ESI and partners 2021)):

- Solutions are needed for MBSE to pay off in a brownfield situation (legacy systems) Solutions are needed for MBSE to support platform-based R&D
- Solutions are needed for MBSE to support managing a large system of diversity
- Solutions are needed for MBSE to multi-disciplinarily model system qualities
- Solutions are needed to combine MBSE with agile R&D approaches

Most organizations now innovate by leveraging their network partners' added value (such as supply chain partners, innovation partners, and service partners). Thus, methods and tools should be able to share and use models as the authoritative source of systems engineering information *across* organizational boundaries, ecosystems, and supply chains. As parties are typically involved in multiple networks, they are expected to handle models originating from multiple MBSE methods and tools.

ESI and Dutch high-tech equipment industry partners collaborate to explore these needs together. This Dutch ecosystem has the unique advantage of not being competitors, yet still grappling with a similar type of systems engineering issues. The first phase of this study has identified needs, drivers, and influencing factors and created guidance as a reasoning line. Further in-depth workshops will elaborate on selected topics and deepen this reasoning line in the coming period.

ESI is also interested in organizing further exchanges and wider experience sharing, including across industry sectors, in elaborating and sharpening insights on adding value with the introduction of MBSE in organizations. ■

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